

TIANYU LU

[Phone TBD] · [Email TBD]

EDUCATION

Texas Academy of Mathematics and Science

University of North Texas | Denton, TX Aug. 2025-May 2027 (Expected)
Residential early-college program; university coursework in Grades 11-12.
Current GPA: 3.889/4.00

Jasper High School | Plano, TX

Aug. 2023-May 2025

Completed Grades 9-10 before entering TAMS.

ACADEMIC & ENGINEERING ACTIVITIES

Student Researcher | University of North Texas, Prof. Qing Yang's Lab

11-12

15 hrs/wk, 44 wks/yr

- Developed an attention-enhanced proximal policy optimization model to dispatch unmanned ground vehicles using aerial sensing information from a UAV.
- Studied task allocation first in an Open-RMF logistics environment, then adapted the same coordination problem to a search-and-rescue setting in which a UAV locates targets and ground vehicles respond.
- Implemented and evaluated scheduling simulations, analyzed results, communicated progress with the research team, and revised the project in response to feasibility and modeling constraints.
- Prepared a first-author manuscript submitted on Aug. 1, 2026 to the IEEE Annual Congress on Artificial Intelligence of Things; manuscript remains under review.

Commented [1]: achieved lowest completion time & queue variance in tests? 这个活动列表中有，统一一下加到这边那个位置合适？

Hardware and Flight Lead | EkDrone Racing UAV Project

11-12

25 hrs/wk, 10 wks/yr

- Built a racing drone with three teammates and took primary responsibility for integrating hardware, checking component compatibility, and tuning receiver and remote-control parameters.
- Adjusted the power-to-weight balance and flight parameters for high-speed performance; incorporated specialized racing-drone materials and 3D-printed components informed by fluid-dynamics simulation.
- Created and completed a pre-flight risk checklist, managed flight operations, and conducted the prototype's maiden flight after the team verified the system.

Research Assistant | UNT Autonomous-Driving Perception Project

11-12

5 hrs/wk, 44 wks/yr

- Used MMDetection as the foundation for a training pipeline that combined Transformer networks with occupancy modeling to interpret multi-camera visual data.
- Worked on locating nearby vehicles relative to the ego vehicle and compared the complexity and stability of six-view training with single-view inputs.
- Outlined a next-stage extension from perception to motion planning and trajectory prediction.

Organizer | Wing Washout Aerodynamics Study

10

10 hrs/wk, 8 wks/yr

- Investigated how wing washout could delay wing-tip stall while introducing an efficiency trade-off in a simplified commercial-aircraft wing model for increasing commercial aircraft efficiency and safety.
- Built and meshed the model in ANSYS Fluent, compared multiple washout and angle-of-attack cases, and used the simulations to identify a promising design range rather than claim a universal aircraft solution.
- Troubleshot negative-volume errors through local mesh refinement and boundary-layer settings; used the SST k-omega turbulence model to improve treatment of vortical flow.

- Presented the simulation-based project at the school Science Fair and documented the need to validate numerical results against appropriate reference or experimental data.

Team Researcher | *UT Dallas STEM Bridge Summer Research Camp* 10
15 hrs/wk, 6 wks

- Worked with a team under Dr. Jie Zhang on machine-learning methods for wind-turbine condition monitoring using high-resolution electrical signals.
- Helped prototype a low-cost, data-driven condition-monitoring approach focused on identifying generator faults, reducing unplanned downtime, and supporting maintenance decisions.
- Maximizes turbine lifetime, reduces downtime, and boosts efficiency by targeting high maintenance costs (38% of operating expenses) and generator failures (17% of turbine failures).
- Connected model performance to operational constraints and presented the project; the team received the program's Best Science Education Award.

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Engineering Project Team Member | *School Marble-Sorter Project* 10
8 hrs/wk, 4 wks

- Built a system to distinguish and route wood, glass, and metal marbles using observable material and color differences.
- Diagnosed unreliable glass-marble detection caused by similar colors and added blue tape at the exit so the sensor could register glass marbles consistently as they passed.

LEADERSHIP, COMMUNICATION & SERVICE

Founder, Host and Producer | *Ximalaya Audio Channel and Engineering Podcasts* 2018-Present
2 hrs/wk, 30 wks/yr

- Began publishing recitations and English dubbing in 2018, building long-term practice in vocal delivery, pacing, recording, and revision.
- Relaunched the channel in late 2025 with 23 original episodes: 15 episodes of Engineering Insights and 8 episodes of Self-Driving Beginner's Journal.
- Researched documented engineering accidents, reconstructed the technical decisions behind them, and translated complex failure mechanisms into scripts for a general Chinese-speaking audience.
- Recorded learning notes about autonomous driving, including corrections to earlier misunderstandings; the combined programs have received 16.8k plays.

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Founder and President | *Jasper High School Physics Club* 10
2 hrs/wk, 16 wks/yr

- Founded the school's first physics club and recruited more than 60 members.
- Planned recurring workshops on college-level physics topics and demonstrations, prepared explanations and practice problems, and coordinated member participation.
- Formed and coached a Physics Bowl team that earned second place in Texas; transferred club responsibilities to a successor when leaving Jasper for TAMS.

Physics Competition Committee Head | *Mu Alpha Theta, TAMS* 2026-Present
2 hrs/wk, 16 wks/yr

- Leads weekly mechanics sessions and works through past $F=ma$ problems with students preparing for physics competitions.
- Assembled and organized a reusable problem bank from prior examinations to support structured practice and discussion.

Volunteer Mathematics Tutor | *Stephens Elementary School* 11-12
1.5 hrs/wk, 10 wks/yr

- Tutors 20 fourth- and fifth-grade students in mathematics through weekly small-group instruction and problem-solving practice.

- Organized MathFun, a game-based session designed to draw students into mathematical reasoning through accessible challenges.

Fundraising and Service Volunteer | *Alliance of Youth Leaders in the United States* 9-10
1 hr/wk, 16 wks/yr

- Operated concession stands at community parks and personally raised approximately \$300 for programs serving children with disabilities.
- Packed donated books for communities in Africa and prepared meal packages through Feed My Starving Children.

ATHLETICS

Point Guard and Independent Strength Trainee | *TAMS Basketball Team* 2023-Present
5 hrs/wk, 40 wks/yr

- Played as point guard in the UNT Open Basketball Tournament and participated in regular team practices and games.
- Maintains a near-daily independent routine combining strength training and running, continuing a habit established in middle school.

Player | **TAMS Table Tennis Team** 2023-Present
2 hrs/wk, 16 wks/yr

- Compete weekly in club singles & doubles matches; drill forehand/backhand footwork and technique .

Bridge Player 2014-2023
2 hrs/wk, 40 wks/yr

- **Competed in** tournament bridge from Grade 1 through Grade 9 (2014–2023).
- Learned bidding systems, card-play strategy & partnership communication through regular training and local tournaments.
- Coordinated tactics with partner under incomplete information; inferred opponents' hands from limited clues and weighed risk vs. reward in real-time team decisions.
- **Ran** structured post-game reviews to locate errors and iterate strategy.

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AWARDS & HONORS

- UNT Undergraduate Research Fellowship (Grade 11-12)
- USA Mathematical Olympiad (USAMO) Qualifier; AIME best score: 12/15 (Grades 11)
- AMC 10 Distinguished Honor Roll, top 1%; AMC 12 Distinction, top 5% (Grades 10-11)
- USA Physics Olympiad (USAPhO) Qualifier via the F=ma Exam (Grade 10)
- Physics Bowl, 2nd-Place Team in Texas, Region 11, Division 1 (Grade 10)
- Plano ISD Superintendent's Scholar Award, district-wide (Grade 10)
- Best Science Education Award, Texas Association of Science and Technology STEM Bridge Summer Research Camp (Grade 10)

TECHNICAL SKILLS

Autonomous systems and machine learning: PPO-based task dispatching; Open-RMF; MMDetection; Transformer and occupancy-model workflows.

Drone hardware and control system: Drone hardware integration and flight tuning; 3D print; pre-flight risk checks.

Mechanical modeling and simulating: ANSYS Fluent modeling and meshing; aerodynamic simulation.

Communication: Research writing; technical podcast scripting, recording, and production.

Program Languages: Python, Java, C++.